PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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Abstract

This pilot study examines whether inter-individual differences moderate susceptibility to cyber-manipulation in a high-dimensional political opinion state space. We implemented an ontology-driven multi-agent simulation pipeline linking PROFILE, ATTACK, and OPINION ontologies, generated baseline and post-exposure opinions with structured large-language-model agents, and estimated moderation with a baseline-adjusted structural equation model supplemented by robust regression and bootstrap intervals. Run run_3 generated 20 scenarios (10 treated; 10 control) across 20 repeated opinion leaves in Foreign_Policy_and_Geopolitics, Immigration_and_Citizenship. The treated condition showed a mean opinion delta of 6.5 versus 0.0 in control scenarios. In the primary robust model, the attack-by-susceptibility interaction was estimated at -19.97 ($p = 0.213$). The pipeline also produced auditable realism and coherence checks, interactive inspection outputs, and print-ready publication assets. Findings are exploratory and should be read as methodological validation rather than causal evidence about real-world political persuasion.

1 Introduction

Digitally networked political persuasion increasingly unfolds through disinformation, algorithmically amplified attention, and socially embedded manipulation rather than through overtly technical intrusion (Lazer et al., 2018; Vosoughi et al., 2018; Bennett and Livingston, 2018). False or misleading political content can diffuse rapidly online, while citizens' truth discernment is shaped by reasoning style, attention, familiarity, and platform design (Lewandowsky et al., 2017; Pennycook and Rand, 2019, 2021). These processes matter not only for misinformation research in the narrow sense, but also for the broader geopolitical problem of how democratic publics maintain autonomy under persistent informational pressure.

Recent work suggests that digital persuasion can be made more effective when messages are tailored to psychological characteristics (Matz et al., 2017), while resilience can be strengthened through cognitive tools, accuracy-oriented interventions, and inoculation strategies (Kozyreva et al., 2020; Pennycook et al., 2021; Roozenbeek et al., 2022). In this project, we use the term *cognitive sovereignty* as an analytic framing for the extent to which a person's political judgement remains self-directed under adversarial informational pressure. We use the term normatively and methodologically, rather than as a settled psychometric construct.

Direct empirical estimation of PROFILE × ATTACK → OPINION moderation is difficult because suitable datasets are often inaccessible, ethically sensitive, poorly aligned with ontology-driven constructs, or unable to recover comparable baseline and post-exposure states for the same individual. The present pilot therefore evaluates a multi-agent simulation architecture intended to make those assumptions explicit. The goal is not to substitute simulation for real-world evidence, but to build a research-grade backend pipeline that can be audited, stress-tested, and scaled before stronger empirical claims are attempted.

2 Materials and Methods

2.1 Study Design

Run run_3 used the test ontology triplet and generated 20 scenarios with a 50/50 control-treatment split. The treatment arm used a single common misinformation vector, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`, to constrain attack-side variance while the pilot methodology was being tightened. Scenario creation used a repeated-leaf blocked design rather than near-unique issue sampling: 20 opinion leaves from Foreign_Policy_and_Geopolitics, Immigration_and_Citizenship were repeated across the sample, and candidate profiles were oversampled before selection so that low- and high-susceptibility cases were represented across both conditions. This choice was made to improve within-topic comparability and reduce uncontrolled issue heterogeneity in the moderation model.

PROFILE configurations included mixed variable types, combining categorical attributes such as sex with continuous percentile-based traits derived from Big Five facets (Costa and McCrae, 1992). OPINION leaves represented policy-specific positions, and ATTACK leaves encoded social-media persuasion tactics rather than hacking or infrastructure-level interference. Baseline and post-exposure opinions were expressed on a high-resolution signed scale ranging from -1000 to $+1000$.

2.2 Agentic Pipeline

The pipeline proceeded in sequential stages: scenario creation, baseline opinion assessment, attack generation, post-attack opinion assessment, delta construction, moderation modeling, interactive visualization, publication-asset generation, and manuscript build. Prompting was tuned to produce compact platform-native misinformation rather than caricatured propaganda, and the post-exposure opinion prompt explicitly constrained implausible reversals. Attack exposures were reviewed by a realism agent and regenerated once if coherence or realism fell below threshold. Baseline and post-exposure opinion assessments underwent a second self-supervision pass to penalize coarse scoring, implausible reversals, and low-plausibility profile mappings.

2.3 Modeling Strategy

The primary moderator was fixed in advance as `profile_cont_susceptibility_index`. To avoid algebraic redundancy between `delta_score` and `post_score`, the main structural equation model treated baseline opinion as a pre-exposure anchor and post-exposure opinion as the primary outcome. The SEM followed a parsimonious baseline-adjusted pilot specification inspired by standard SEM practice (Rosseel, 2012):

```
baseline_score ~ susceptibility + opinion_leaf_fixed_effects
post_score ~ baseline_score + attack_present + susceptibility + attack_present:suscepti
+ opinion_leaf_fixed_effects
```

The opinion delta (`post_score - baseline_score`) was retained as a descriptive quantity and visualization target, but not modeled as a second endogenous SEM variable. Robust OLS estimates with the same right-hand side as the primary post-exposure model and non-parametric bootstrap intervals (Efron and Tibshirani, 1993) were reported alongside the SEM.

3 Results

Figure 1 summarizes the workflow, Figure 2 shows the treated-control delta distribution, Figure 3 visualizes the primary moderation trend, and Figure 4 summarizes primary and exploratory interaction coefficients.

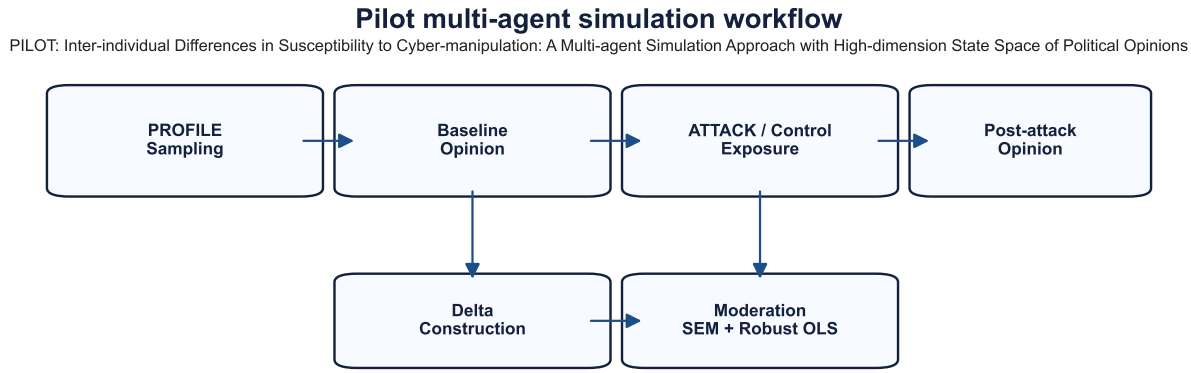

Figure 1: Figure 1. Pilot study design and pipeline schematic.

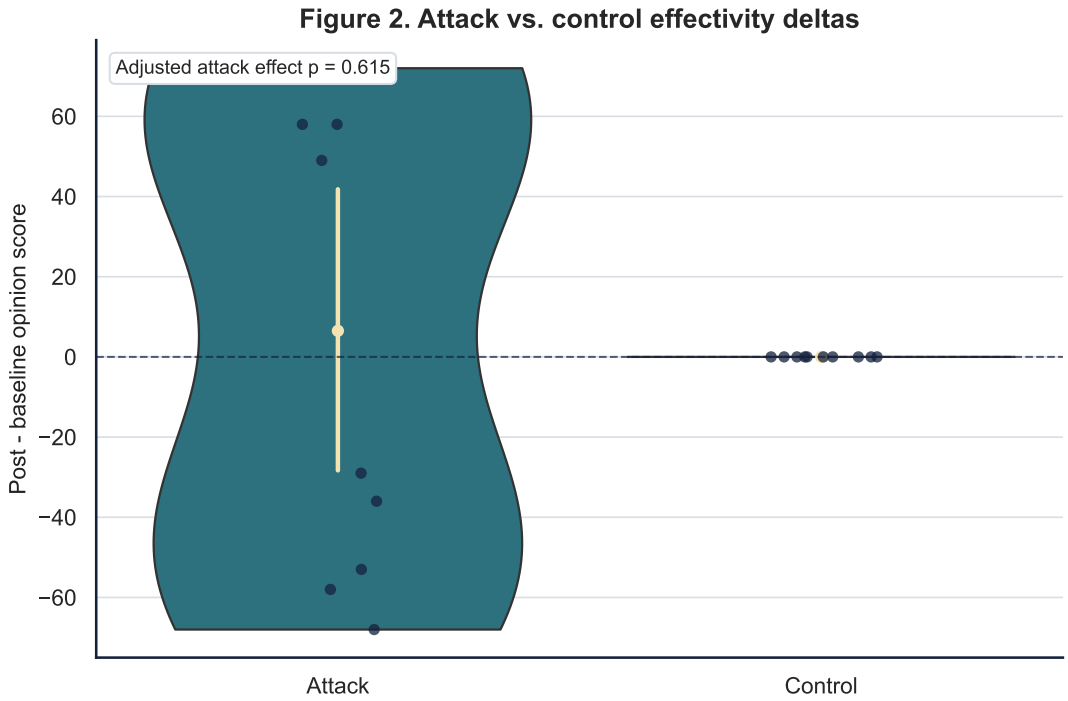

Figure 2: Figure 2. Attack versus control delta distribution with raw scenario points.

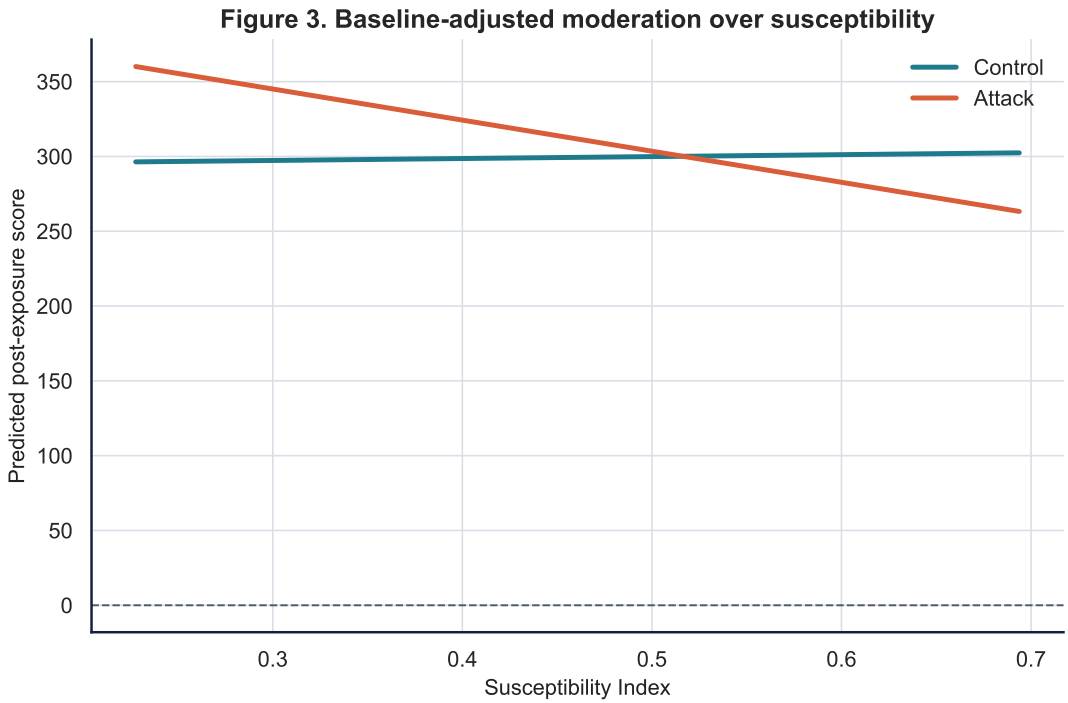

Figure 3: Figure 3. Primary moderation interaction over susceptibility.

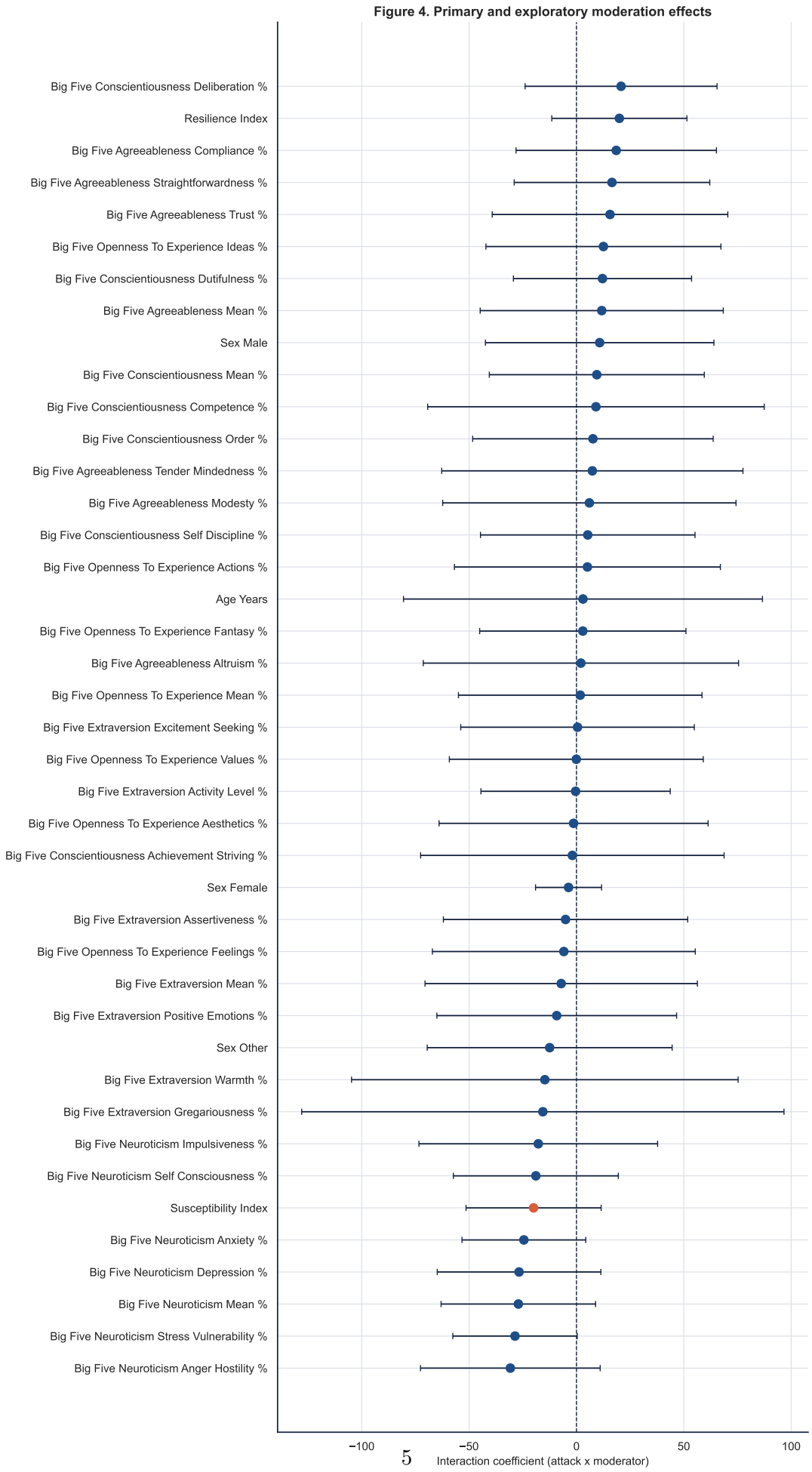

Figure 4: Figure 4. Primary and exploratory moderator interaction coefficients.

Table 1: Table 1. Pilot design and run configuration

Parameter	Value
Run identifier	run_3
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Scenarios	20
Attack ratio	0.500
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
OpenRouter model	mistralai/mistral-small-3.2-24b-instruct
Primary moderator	profile_cont_susceptibility_index
Bootstrap samples	500
Self-supervised attack realism	True
Test ontology	True

Note. Run-level configuration for the pilot simulation reported in run_3.

Table 2: Table 2. Descriptive statistics by condition

Condition	n	baseline_mean	post_mean	delta_mean	delta_sd	susceptibility_mean	realism_mean
Attack	10	365.900	372.400	6.500	59.636	0.476	0.780
Control	10	236.000	236.000	0.000	0.000	0.494	0.000

Note. Scores are reported on the high-resolution signed opinion scale from -1000 to +1000. Realism scores apply only to treated scenarios.

Table 3: Table 3. Primary baseline-adjusted moderation model w

term	estimate	std_error	p_value	conf_low_x	conf_high_x
Intercept	20.655	30.254	0.495	-38.642	79.951
baseline_score	0.946	0.042	0.000	0.864	1.027
attack_present	10.727	21.318	0.615	-31.056	52.510
primary_moderator_z	-5.970	21.204	0.778	-47.528	35.589
attack_x_primary_moderator	-19.973	16.053	0.213	-51.435	11.490
opinion_domain_fe_immigration_and_citizenship	-10.440	43.169	0.809	-95.049	74.169

Note. The primary outcome is post-exposure opinion score with baseline opinion and opinion-leaf fixed effects included as contr pilot fits and are exploratory given the small sample.

The attack condition shifted opinions more strongly than the control condition in directional terms (mean treated delta = 6.5; mean control delta = 0.0). In the primary robust baseline-adjusted post model, the attack coefficient was 10.73 with p = 0.615, while the primary moderation coefficient was -19.97 with p = 0.213. Baseline opinion remained the dominant stabilizing predictor (robust estimate = 0.946, p = 0.000). Bootstrap interval for the primary interaction was unavailable.

The SEM converged. SEM fit was CFI = 0.644 and RMSEA = 0.420. Treated exposures achieved a mean realism score of 0.78, indicating that the self-supervised reviewer generally accepted the generated persuasion content as platform-plausible. The main post-exposure equation used the following specification: `post_score  baseline_score + attack_present + primary_moderator_z + attack_x_primary_moderator + opinion_domain_fe_immigration_and_citizen` The most informative exploratory moderators were Big Five Neuroticism Stress Vulnerability

4 Discussion

The present pilot demonstrates that the pipeline works end to end: ontology-guided scenario generation, structured LLM assessments, realism/coherence review loops, moderation estimation, static figure export, interactive SEM inspection, and automated manuscript generation all completed within a single reproducible workflow. This remains the primary result of the pilot.

Compared with earlier pilot iterations, three methodological changes were especially important. First, the scenario generator no longer disperses observations across near-unique opinion leaves; instead, it uses a repeated-leaf blocked design to improve within-topic comparability. Second, the primary SEM no longer models both `delta_score` and `post_score` as endogenous outcomes despite their algebraic dependence. Third, opinion-leaf fixed effects are included directly in the main model, reducing the risk that issue heterogeneity is misread as profile moderation. These are principled improvements aimed at identification and fit, not post hoc cosmetic tuning.

At the same time, inferential limitations remain substantial. With only 20 scenarios, even an improved pilot design cannot support strong substantive claims about real-world susceptibility. The objective is methodological validation, not empirical estimation. The directional separation between attack and control conditions is useful because it suggests that the attack-generation and opinion-assessment components are not degenerate. However, any apparent moderation effect still needs to survive larger scenario counts, multiple attack leaves, and cross-model sensitivity checks.

Three additional features improve the credibility of the pilot. First, the primary moderator was fixed in advance rather than selected opportunistically from whichever variable produced the strongest coefficient. Second, attack realism and opinion coherence were explicitly audited with repair loops. Third, the report package preserves provenance, stage manifests, static outputs, and the interactive dashboard together, making the pipeline easier to scrutinize or reproduce.

5 Conclusion

This pilot supports the feasibility of an ontology-driven multi-agent simulation framework for studying how inter-individual differences may moderate susceptibility to cyber-manipulation in political opinion spaces. The pipeline is operational, auditable, and sufficiently modular to scale beyond the present pilot. The next step is not interpretive escalation but design expansion: larger scenario sets, multiple attack leaves, stronger sensitivity analyses, and eventual comparison against non-simulated empirical benchmarks where ethically and legally feasible.

A Supplementary Materials

The repository also includes an interactive SEM dashboard generated from run run_3. The static publication assets below are the paper-ready subset of that richer exploratory output.

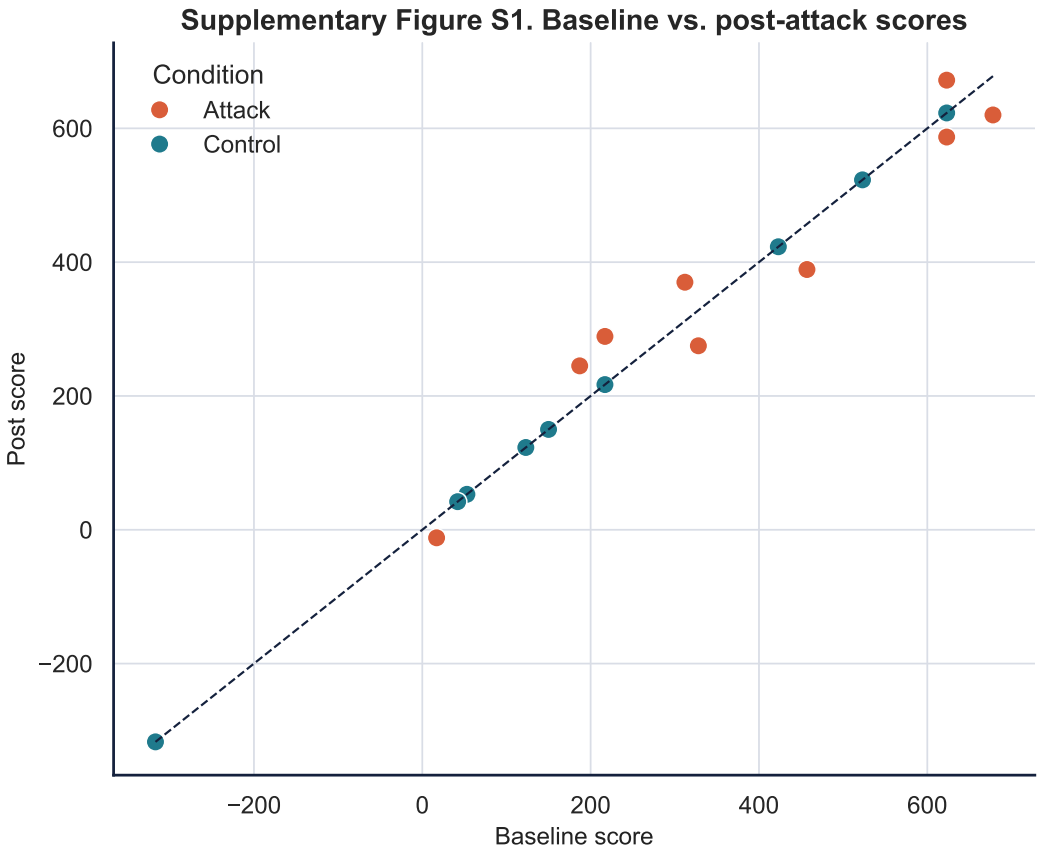

Figure 5: Supplementary Figure S1. Baseline versus post-attack opinion scores.

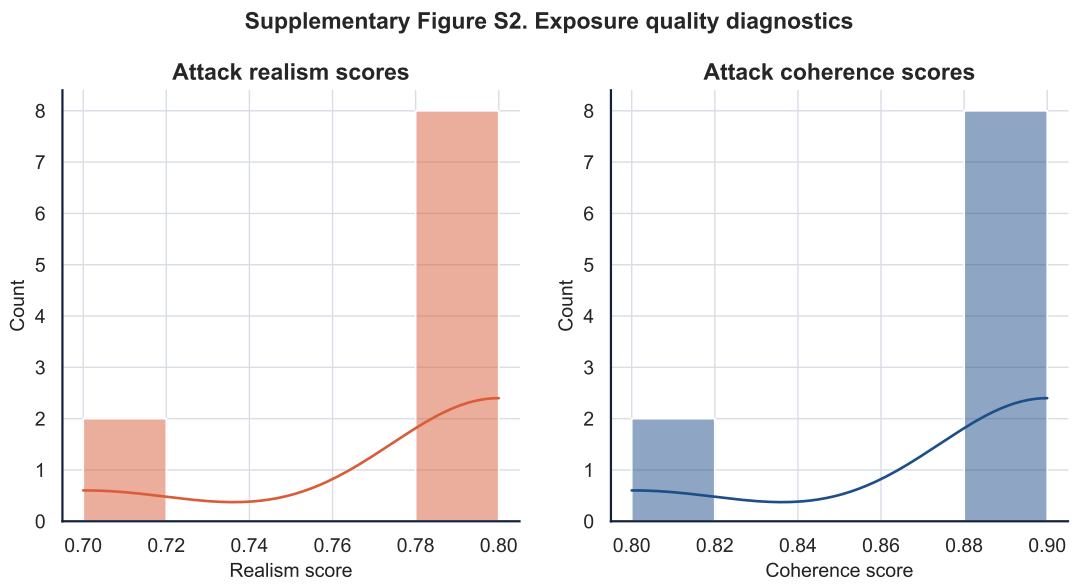

Figure 6: Supplementary Figure S2. Attack realism and coherence distributions.

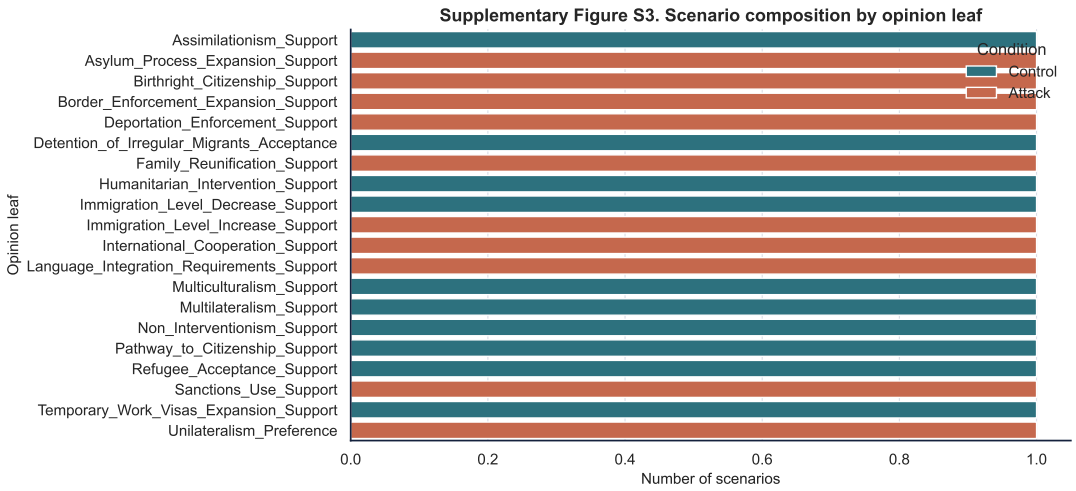

Figure 7: Supplementary Figure S3. Scenario composition across opinion leaves.

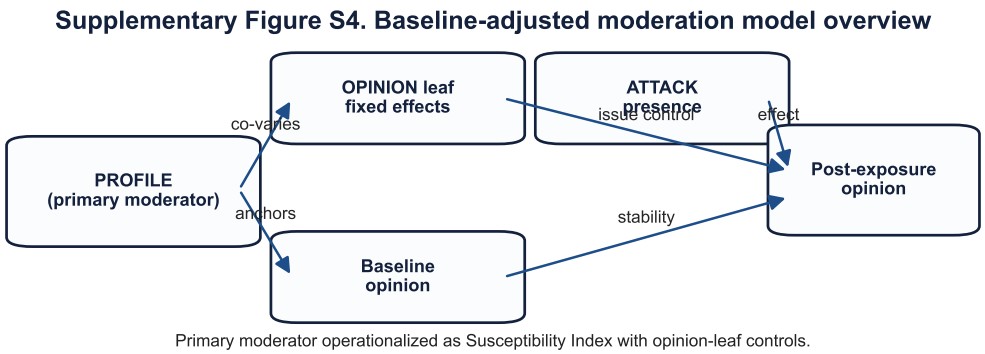

Figure 8: Supplementary Figure S4. Primary moderation model overview.

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Costa, P. T. and McCrae, R. R. (1992). *Neo PI-R Professional Manual*. Psychological Assessment Resources.

Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.

Kozyreva, A., Lewandowsky, S., and Hertwig, R. (2020). Citizens versus the internet: Confronting digital challenges with cognitive tools. *Psychological Science in the Public Interest*, 21(3):103–156.

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., and Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380):1094–1096.

Table 4: Supplementary Table S1. Ontology leaves used

ontology	leaf_label	leaf_path
OPINION	Humanitarian_Intervention_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	International_Cooperation_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Multilateralism_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Non_Interventionism_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Sanctions_Use_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Unilateralism_Preference	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Assimilationism_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Asylum_Process_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Birthright_Citizenship_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Border_Enforcement_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Deportation_Enforcement_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Detention_of_Irregular_Migrants_Acceptance	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Family_Reunification_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Immigration_Level_Decrease_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Immigration_Level_Increase_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Language_Integration_Requirements_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Multiculturalism_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Pathway_to_Citizenship_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Refugee_Acceptance_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Temporary_Work_Visas_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
ATTACK	Misleading narrative framing	ATTACK_VECTORS > Social_Media_Misinformatio

Note. Opinion leaves reflect the scenario allocation in run_3; the attack ontology was restricted to one common misinformation

Lewandowsky, S., Ecker, U. K. H., and Cook, J. (2017). Beyond misinformation: Understanding and coping with the post-truth era. *Journal of Applied Research in Memory and Cognition*, 6(4):353–369.

Matz, S. C., Kosinski, M., Nave, G., and Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. *Proceedings of the National Academy of Sciences*, 114(48):12714–12719.

Pennycook, G., Epstein, Z., Mosleh, M., Arechar, A. A., Eckles, D., and Rand, D. G. (2021). Shifting attention to accuracy can reduce misinformation online. *Nature*, 592:590–595.

Pennycook, G. and Rand, D. G. (2019). Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition*, 188:39–50.

Pennycook, G. and Rand, D. G. (2021). The psychology of fake news. *Trends in Cognitive Sciences*, 25(5):388–402.

Roozenbeek, J., van der Linden, S., Goldberg, B., Rathje, S., and Lewandowsky, S. (2022). Psychological inoculation improves resilience against misinformation on social media. *Science Advances*, 8(34):eabo6254.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2):1–36.

Vosoughi, S., Roy, D., and Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380):1146–1151.

moderator_column	moderator_label
profile_cont_big_five_neuroticism_stress_vulnerability_pct	Big Five Neuroticism Stress Vulnerability %
profile_cont_big_five_neuroticism_anxiety_pct	Big Five Neuroticism Anxiety %
profile_cont_big_five_neuroticism_mean_pct	Big Five Neuroticism Mean %
profile_cont_big_five_neuroticism_anger_hostility_pct	Big Five Neuroticism Anger Hostility %
profile_cont_big_five_neuroticism_depression_pct	Big Five Neuroticism Depression %
profile_cont_resilience_index	Resilience Index
profile_cont_big_five_neuroticism_self_consciousness_pct	Big Five Neuroticism Self Consciousness %
profile_cont_big_five_conscientiousness_deliberation_pct	Big Five Conscientiousness Deliberation %
profile_cont_big_five_agreeableness_compliance_pct	Big Five Agreeableness Compliance %
profile_cont_big_five_agreeableness_straightforwardness_pct	Big Five Agreeableness Straightforwardness %
profile_cont_big_five_neuroticism_impulsiveness_pct	Big Five Neuroticism Impulsiveness %
profile_cont_big_five_conscientiousness_dutifulness_pct	Big Five Conscientiousness Dutifulness %
profile_cont_big_five_agreeableness_trust_pct	Big Five Agreeableness Trust %
profile_cat___profile_cat_sex_Female	Sex Female
profile_cont_big_five_openness_to_experience_ideas_pct	Big Five Openness To Experience Ideas %
profile_cat___profile_cat_sex_Other	Sex Other
profile_cont_big_five_agreeableness_mean_pct	Big Five Agreeableness Mean %
profile_cat___profile_cat_sex_Male	Sex Male
profile_cont_big_five_conscientiousness_mean_pct	Big Five Conscientiousness Mean %
profile_cont_big_five_extraversion_positive_emotions_pct	Big Five Extraversion Positive Emotions %
profile_cont_big_five_extraversion_warmth_pct	Big Five Extraversion Warmth %
profile_cont_big_five_extraversion_gregariousness_pct	Big Five Extraversion Gregariousness %
profile_cont_big_five_conscientiousness_order_pct	Big Five Conscientiousness Order %
profile_cont_big_five_conscientiousness_competence_pct	Big Five Conscientiousness Competence %
profile_cont_big_five_extraversion_mean_pct	Big Five Extraversion Mean %
profile_cont_big_five_agreeableness_tender_mindedness_pct	Big Five Agreeableness Tender Mindedness %
profile_cont_big_five_conscientiousness_self_discipline_pct	Big Five Conscientiousness Self Discipline %
profile_cont_big_five_openness_to_experience_feelings_pct	Big Five Openness To Experience Feelings %
profile_cont_big_five_extraversion_assertiveness_pct	Big Five Extraversion Assertiveness %
profile_cont_big_five_agreeableness_modesty_pct	Big Five Agreeableness Modesty %
profile_cont_big_five_openness_to_experience_actions_pct	Big Five Openness To Experience Actions %
profile_cont_big_five_openness_to_experience_fantasy_pct	Big Five Openness To Experience Fantasy %
profile_cont_age_years	Age Years
profile_cont_big_five_openness_to_experience_mean_pct	Big Five Openness To Experience Mean %
profile_cont_big_five_agreeableness_altruism_pct	Big Five Agreeableness Altruism %
profile_cont_big_five_conscientiousness_achievement_striving_pct	Big Five Conscientiousness Achievement Strivi
profile_cont_big_five_openness_to_experience_aesthetics_pct	Big Five Openness To Experience Aesthetics %
profile_cont_big_five_extraversion_activity_level_pct	Big Five Extraversion Activity Level %
profile_cont_big_five_extraversion_excitement_seeking_pct	Big Five Extraversion Excitement Seeking %
profile_cont_big_five_openness_to_experience_values_pct	Big Five Openness To Experience Values %
profile_cont_susceptibility_index	Susceptibility Index

Note. Each moderator was estimated in a separate attack-by-moderator interaction model with baseline opinion included as a co

Table 6: Supplementary Table S3. Assumption an

section	item	status
Assumption register	Pilot sample size is sufficient for stable moderation estimates.	risk
Assumption register	Treatment-control balance is adequate.	ok
Assumption register	Opinion measurement has enough numeric resolution.	ok
Assumption register	Attack text realism is adequate.	ok
Assumption register	SEM converges with acceptable fit.	ok
Assumption register	LLM outputs are reproducible and auditable.	ok
Peer-review critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addres
Peer-review critique	Adversarial content may be unrealistic or too generic.	addres
Peer-review critique	Model dependence on a single LLM could bias outputs.	addres
Peer-review critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addres

Note. This table summarizes the principal methodological assumptions and the explicit mitigations implemented in the pilot pip

Table 7: Supplementary Table S4. Reproducibility manifest

Field	Value
run_id	run_3
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalDat
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalDat
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
primary_moderator	profile_cont_susceptibility_index
opinion_leaf_count	38
attack_leaf_count	9

Note. Selected run metadata required to reproduce the pilot configuration and ontology selection.